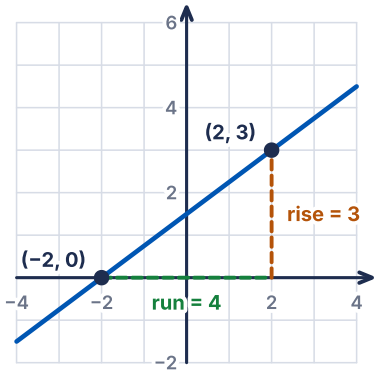
# Understanding Slope

Name _____ Date _____ Period _____

Guided Notes

## What is Slope?

Imagine you're walking up a hill. Some hills are steep, some are gentle. **Slope** is the mathematical way to measure that steepness.




Walk right 4, climb up 3. Slope measures that climb.

**Slope** measures the *rate of change* — how much the y-value changes for every 1 unit change in x. It tells us two things:
**1.** How steep the line is
**2.** Which direction it goes (up or down, left to right)

We use the letter **m** to represent slope. In the equation **y = mx + b**:

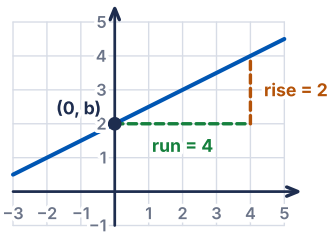
**m** = the slope (steepness and direction)
**b** = the y-intercept (where the line crosses the y-axis)

📝 Write this in your notes: "Slope = rate of change = rise over run = m"

## ✏ Practice: What Is Slope?

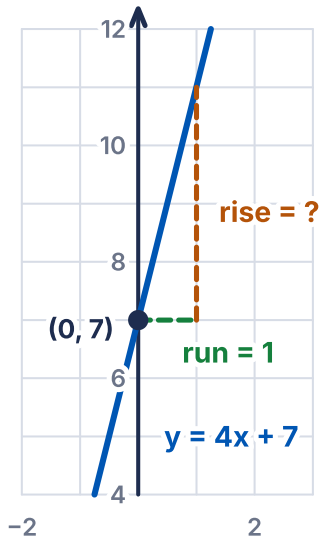
**1.** ★ Try it In **y = mx + b**, what does the letter **m** stand for?

- ○ the slope
- ○ the y-intercept
- ○ the x-value



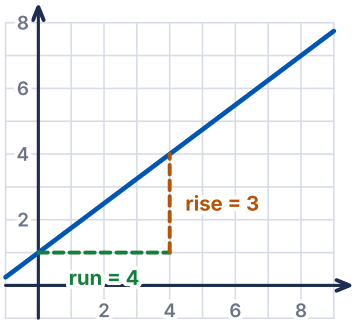
**2.** ★★ On level What is the slope of **y = 4x + 7**? Write just the number.

**Answer:** _____



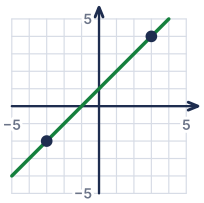

**3.** ★★★ Challenge This hill goes right 4 and up 3. If you walk right 8 on the same hill, how far up do you climb?

**Answer:** _____



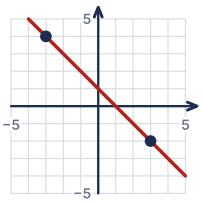
## Four Types of Slope

There are exactly **four types** of slope:



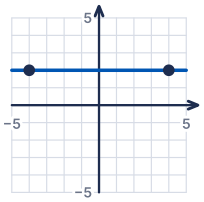

**Positive ↗**

m > 0



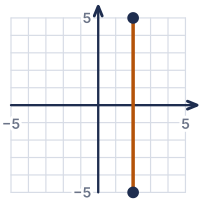

**Negative ↘**

m < 0




**Zero →**

m = 0




**Undefined Slope ↕**

Vertical line.

m = undefined

**Positive slope:** As x increases, y also increases. Think of walking *uphill*.

**Negative slope:** As x increases, y decreases. Think of walking *downhill*.

**Zero slope:** A perfectly flat, horizontal line. No rise at all — the y-value never changes. Think of walking on *flat ground*.

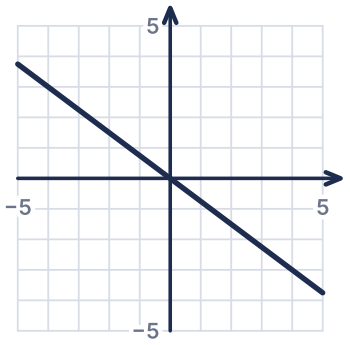
**Undefined slope:** A vertical line. The run is 0, and we can't divide by zero! Think of a *cliff wall* — you can't walk up that.

📝 Memory trick: **"Zero slope = Zero (flat), Undefined = Up and Down (vertical)"**

## ✏ Practice: Four Types of Slope

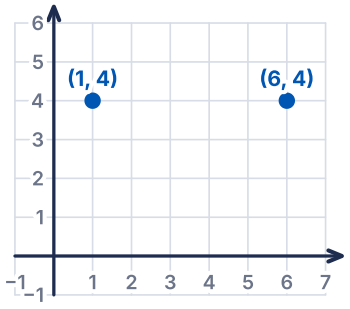
**1.** ★ Try it What type of slope does this line have?

- ○ Positive
- ○ Negative
- ○ Zero
- ○ Undefined



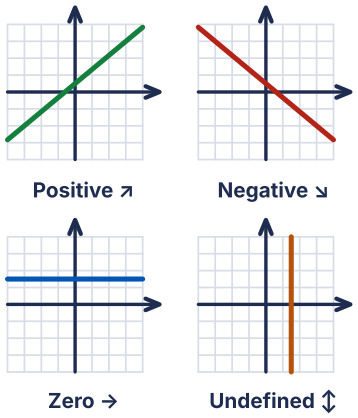

**2.** ★★ On level A line goes through **(1, 4)** and **(6, 4)**. What type of slope does it have?

- ○ Positive
- ○ Negative
- ○ Zero
- ○ Undefined




**3.** ★★★ Challenge Which equation has a **negative** slope?

- ○ y = 3x − 5
- ○ y = −2x + 4
- ○ y = x + 1
- ○ y = 5x



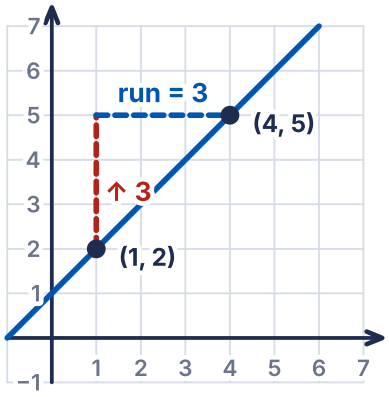

### Finding Slope from a Graph: Rise Over Run

To find slope from a graph, pick **any two points** on the line, then count:

$$\text{Slope} = \frac{\text{rise (vertical change)}}{\text{run (horizontal change)}}$$

**Rise** = how far you move **up or down** (count vertical grid squares)
**Run** = how far you move **left or right** (count horizontal grid squares)

### Example 1: Line through (1, 2) and (4, 5)



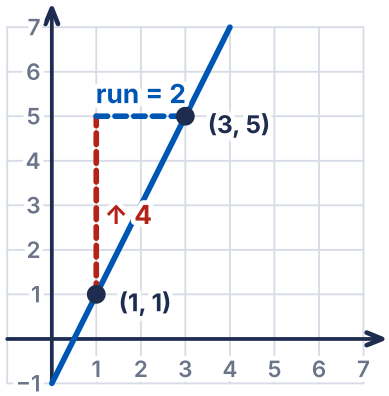

Starting at **(1, 2)**:

→ Count **UP 3** squares (rise = 3)
→ Count **RIGHT 3** squares to reach (4, 5) (run = 3)

> Slope = $\frac{\text{rise}}{\text{run}} = \frac{3}{3} = 1$

### Example 2: A steeper line

Line through **(1, 1)** and **(3, 5)**.

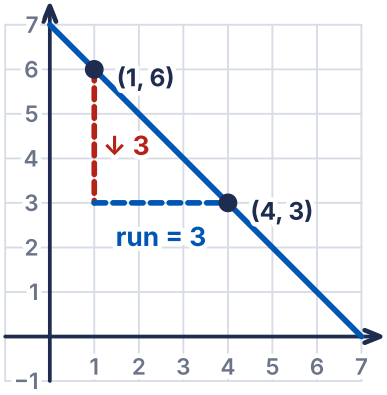



→ **Rise = 4** (count up from 1 to 5)
→ **Run = 2** (count right from 1 to 3)

> Slope = $\frac{4}{2} = 2$

### Example 3: Negative slope (going downhill)

Line through **(1, 6)** and **(4, 3)**.




→ **Rise = −3** (count DOWN from 6 to 3 — negative because you go down!)
→ **Run = 3** (count right from 1 to 4)

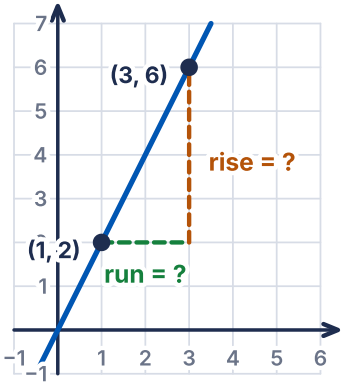
> Slope = $\frac{-3}{3}$ = **−1**

> 📝 If you count **down**, the rise is **negative**. That's what makes the slope negative!

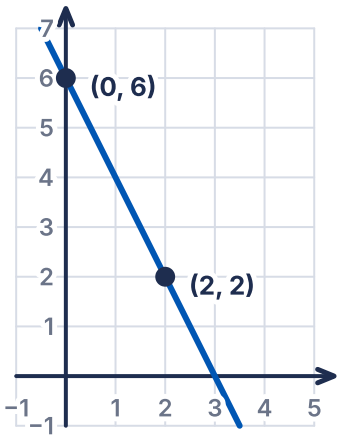
## ✏ Practice: Rise Over Run

**1.** ★ Try it Count the rise and the run from one dot to the other. What is the slope?

**Answer:** _____

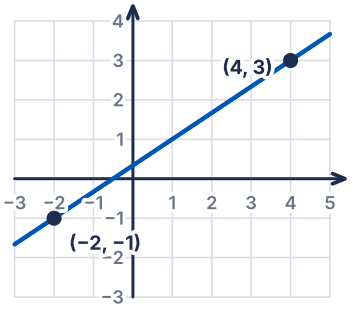


**2.** ★★ On level Find the slope of this line. Use the two dots.




**Answer:** _____

**3.** ★★★ Challenge Find the slope of this line. Write it as a fraction in simplest form, like 3/4.




**Answer:** _____

## The Slope Formula

When you have two points but no graph, use the **slope formula**:

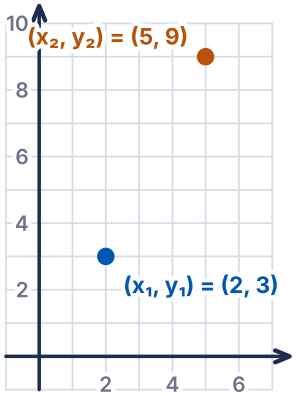
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

> **$(x_1, y_1)$** is the first point
> **$(x_2, y_2)$** is the second point
> The **numerator** (top) is the rise. The **denominator** (bottom) is the run.

**Example 1: Find the slope of the line through (2, 3) and (5, 9)**

**Step 1:** Label your points

> $(x_1, y_1) = (2, 3)$
> $(x_2, y_2) = (5, 9)$



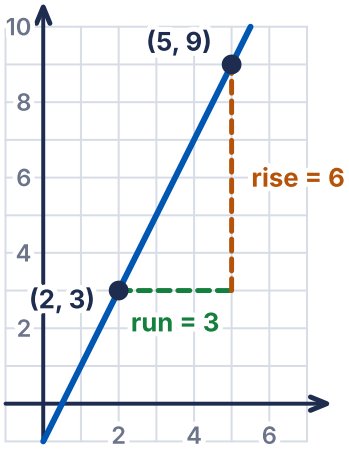

**Example 1 (continued)**

**Step 2:** Substitute into the formula

$$m = \frac{9 - 3}{5 - 2}$$

**Step 3:** Simplify

$$m = \frac{6}{3} = 2$$



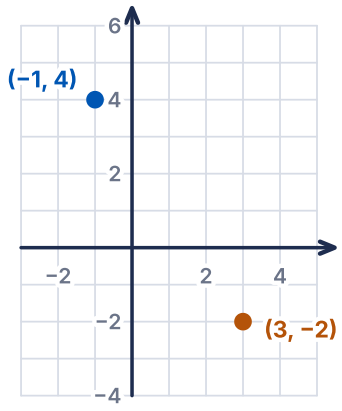

rise ÷ run = 6 ÷ 3 = 2

📝 The slope is **2**, which means for every 1 unit you move right, you move up 2 units.

**Example 2: (−1, 4) and (3, −2)**

**Step 1:** Label your points

$(x_1, y_1) = (-1, 4)$
$(x_2, y_2) = (3, -2)$




**Step 2:** Substitute

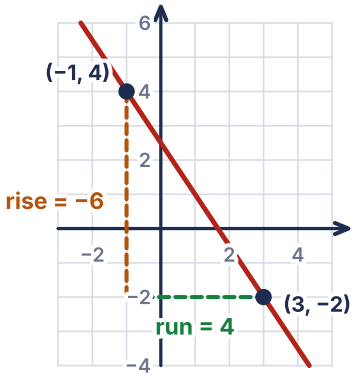
$$m = \frac{-2 - 4}{3 - (-1)}$$

**Example 2 (continued)**

**Step 3:** Simplify carefully — watch the signs!

$$m = \frac{-6}{4} = \frac{-3}{2}$$

> **Important:** Always simplify your fraction! $-\frac{6}{4}$ reduces to $-\frac{3}{2}$.
> The negative sign means the line goes *downhill* (left to right).



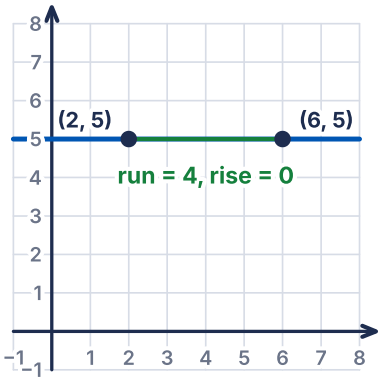

Going DOWN 6 while going right 4: m = −6 ÷ 4 = −3/2

> ⚠️ **Common mistake:** 3 − (−1) = 3 + 1 = 4, NOT 3 − 1 = 2. Subtracting a negative = adding!

**Example 3: (2, 5) and (6, 5)**

$$m = \frac{5 - 5}{6 - 2} = \frac{0}{4} = 0$$

> **Zero in the numerator → slope = 0 → horizontal line!**
> Both y-values are the same, so there's no vertical change.

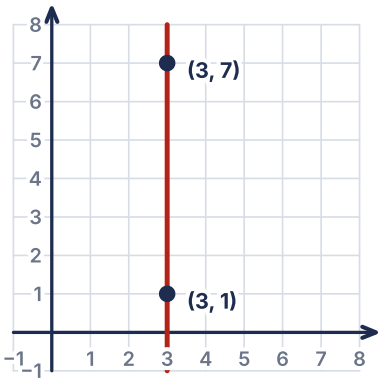



Flat line: rise = 0, so m = 0

**Example 4: (3, 1) and (3, 7)**

$$m = \frac{7 - 1}{3 - 3} = \frac{6}{0} = \text{undefined}$$

**Zero in the denominator → slope is undefined → vertical line!**
Both x-values are the same. Division by zero is undefined.

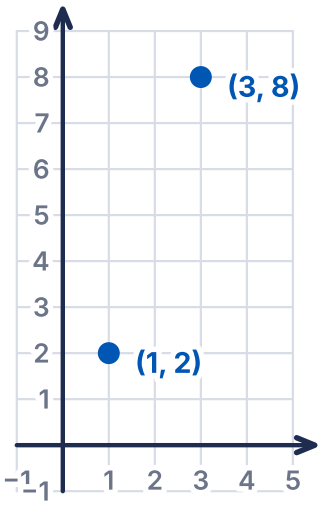



Vertical line: run = 0, and we cannot divide by 0

## ✏ Practice: The Slope Formula

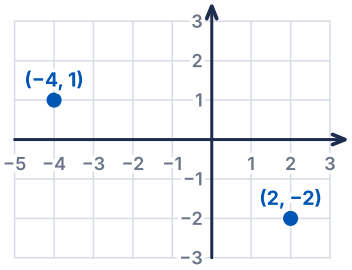
**1.** ★ Try it Use the slope formula to find the slope through **(1, 2)** and **(3, 8)**.

**Answer:** _____



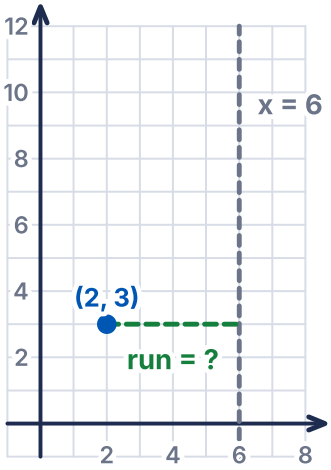

**2.** ★★ On level Find the slope through **(−4, 1)** and **(2, −2)**. Write a fraction in simplest form.

**Answer:** _____




**3.** ★★★ Challenge A line goes through **(2, 3)** and **(6, y)**. Its slope is 2. What is y?

**Answer:** _____




(6, y) is somewhere on the dashed line x = 6.

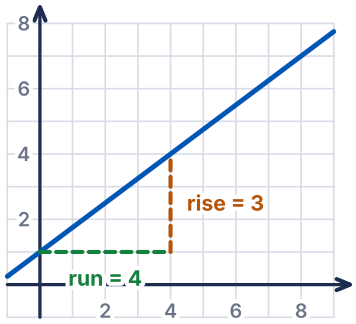
## Comprehensive Practice

Mixed practice from every section: what slope means, the four types, rise over run, and the slope formula. If an answer is a fraction, write it like 3/4 or -1/2.

### ✏ Comprehensive Practice All sections

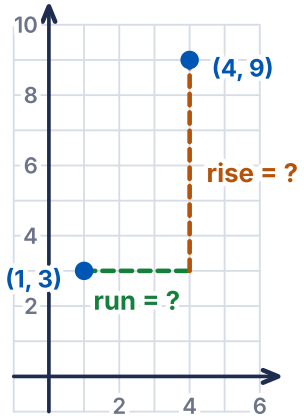
**1.** ★ Try it What does slope measure?

- ○ How steep a line is
- ○ Where a line crosses the y-axis
- ○ How long a line is



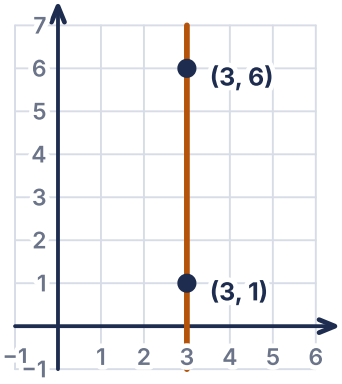

**2.** ★ Try it Find the slope through (1, 3) and (4, 9).

**Answer:** _____




**3.** ★★★ Challenge Jay says, "A vertical line has a slope of 0." Is Jay right?

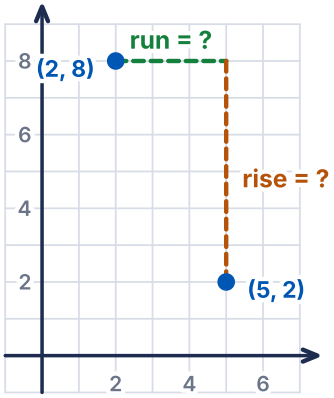
- ○ Yes, a vertical line has slope 0
- ○ No, a vertical line has an undefined slope
- ○ No, a vertical line has a negative slope




A vertical line

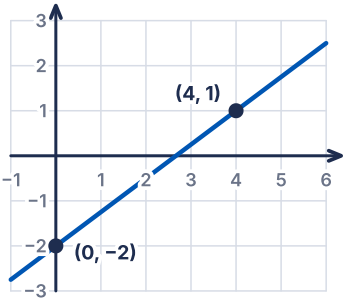
**4.** ★★ On level Find the slope through (2, 8) and (5, 2).

**Answer:** _____



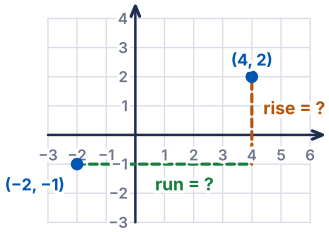

**5.** ★★ On level What is the slope of this line? Write a fraction, like 1/2.

**Answer:** _____



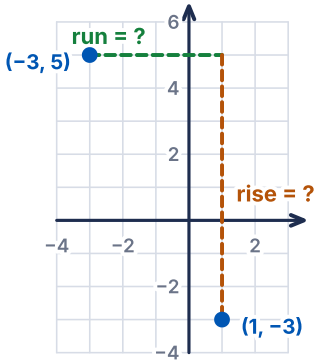

**6.** ★★★ Challenge Find the slope through (−2, −1) and (4, 2). If the answer is a fraction, write it as a/b (like 3/4).

**Answer:** _____

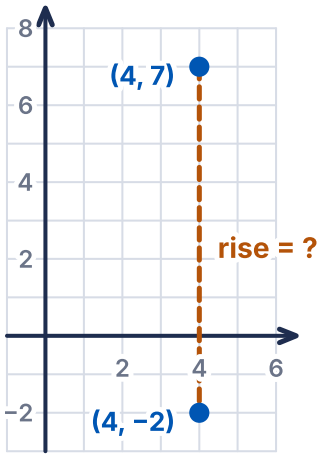



**7.** ★★ On level Find the slope through (−3, 5) and (1, −3).

**Answer:** _____

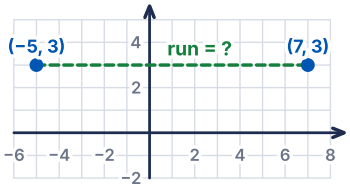


**8.** ★★ On level Find the slope through (4, 7) and (4, −2). Write “undefined” if the slope is undefined.



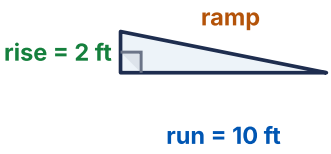

**Answer:** _____

**9.** ★ Try it Find the slope through (−5, 3) and (7, 3).




**Answer:** _____

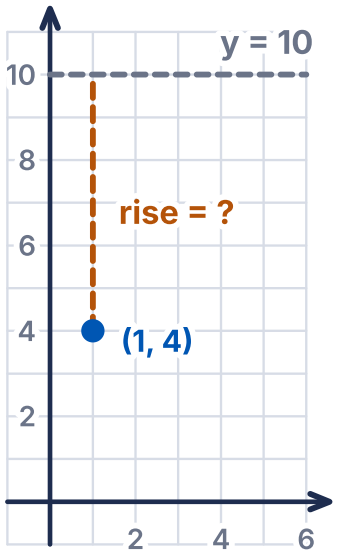
**10.** ★★★ Challenge A ramp rises 2 feet over a run of 10 feet. What is its slope? Write a fraction in simplest form.




**Answer:** _____

**11.** ★★★ Challenge A line goes through **(1, 4)** and **(x, 10)**. Its slope is 3. What is x?

**Answer:** _____




(x, 10) is somewhere on the dashed line y = 10.